SLURM
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DAGSTER

Why orchestration matters

- A raw “engine” (script, notebook, or binary) is not enough once you depend on sensors, ETL, or ML training.
- Orchestrators provide dependency tracking, retries, metrics, and the control plane HPC teams lack.
- Dagster supplies that missing layer; dagster-slurm connects it to the supercomputer’s scheduler.

Asset code stays the same

- Same Dagster asset runs locally or on Slurm; only configuration changes.
- Compute resource wraps payload execution with observability and retries.

```python
import dagster as dg
from dagster_slurm import BashLauncher, ComputeResource

@dg.asset
def train_pytorch(context: dg.AssetExecutionContext, compute: ComputeResource):
    script_path = dg.file_relative_path(
        __file__, "../workloads/pytorch/train_classifier.py"
    )

    completed = compute.run(
        context=context,
        payload_path=script_path,
        launcher=BashLauncher(),
        extra_env={"EPOCHS": "3", "BATCH_SIZE": "128"},
    )
    yield from completed.get_results()
```

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- Compute resource wraps payload execution with observability and retries.

```python
def main():
    context = PipesContext.get()
    num_epochs = context.extras["epochs"]
    context.log.debug(f"Number of epochs: {num_epochs}")
    input_path = os.environ.get("INPUT_DATA")
    context.log.info(f"Input: {input_path}")
    # Your processing logic here (pytorch, ... )
    result = {"rows_processed": 1000}
    context.report_asset_materialization(
        metadata={
            "rows": result["rows_processed"],
            "processing_time": "10s",
        }
    )

if __name__ == "__main__":
    with open_dagster_pipes() as context:
        main()
```

Different strengths, better together

Data orchestrator

- Models business data products and when they need to refresh.
- Captures lineage, metadata, and failures across heterogeneous systems.
- Keeps engineers productive with local runs, tests, and incremental deploys.

Supercomputer + Slurm

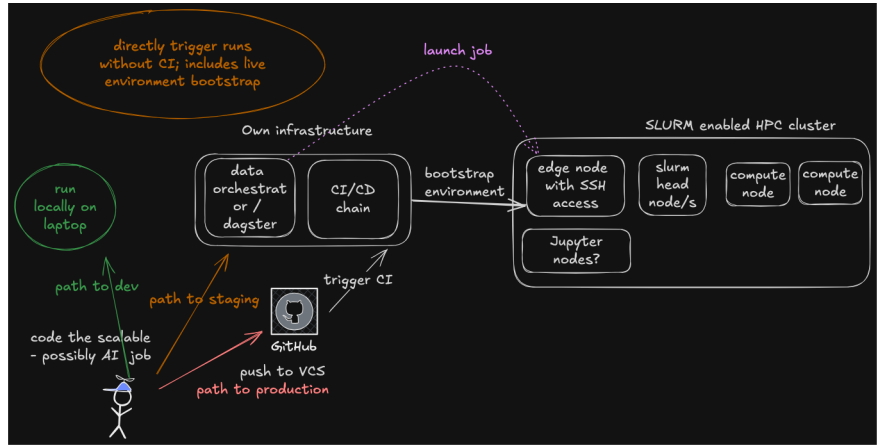
- Maximises utilisation of scarce accelerators and node hours.
- Enforces fair-share policies, job placement, and low-level resource binding.
- Provides high-performance file systems and interconnect-aware scheduling.

Bridge

- dagster-slurm lets Dagster plan the work while Slurm owns the physical execution.
- One orchestrator spans tools silos (HPC and non-HPC) for unified visibility

dagster-slurm in one slide

- **Keep the Dagster control plane** – asset logic, schedules, and policies stay in Dagster regardless of where you run (HPC and non-HPC).
- **Package once, run anywhere** – pixi/pixi-pack builds reproducible environments that deploy to login nodes, edge VMs, or compute partitions.
- **Delegate execution to Slurm** – dagster-slurm translates runs into `sbatch` / `squeue` interactions while respecting quotas and placement rules.
- **See everything in Dagster UI** – queue state, Slurm metrics, and Pipes logs stream back for a single operational console.



EU sovereign GPU cloud does not come out of nowhere

maybe this project can support making HPC systems more accessible

dagster-slurm

Overview **Runs** Assets Automation Deployment

Runs / 50db2417 Success | process_data | Oct 22, 4:14:31 PM | 0:00:07

View asset lineage | View tags and config

Hide not started steps

Re-execute (name:"process_data")

1.4s 2s 3s 4s 5s 6s 7s 8s

process_data

name:"process_data" Hide unselected steps

Preparing (0)

No steps are waiting to execute

Executing (0)

No steps are executing

Errored (0)

Events | stdout | stderr

query:name:"process_data"

Levels (5/6)

TIMESTAMP	OP	EVENT TYPE	INFO
4:14:33.238 PM	process_data	INFO	Starting data processing...
4:14:33.241 PM	process_data	INFO	Input: /path/to/input
4:14:33.244 PM	process_data	INFO	Output: /path/to/output
4:14:33.247 PM	process_data	INFO	Processing complete!
4:14:38.828 PM	process_data	STEP_OUTPUT	Yielded output "result" of type "Any". (Type check passed). rows 1000 processing_time 10s
4:14:38.837 PM	process_data	ASSET_MATERIALI…	Materialized value process_data. asset_key process_data [View Asset] rows 1000 processing_time 10s
4:14:38.858 PM	process_data	ASSET_OBSERVATI…	asset_key process_data [View Asset] deployment_mode local custom_messages_count 0
4:14:38.871 PM	process_data	STEP_SUCCESS	Finished execution of step "process_data" in 6.17s.

Overview Runs Assets Automation Deployment

Assets > process_data

Materialize

Overview Events Checks Lineage Automation

Recent updates

Showing all 2 updates

10/22/2025 10/22/2025

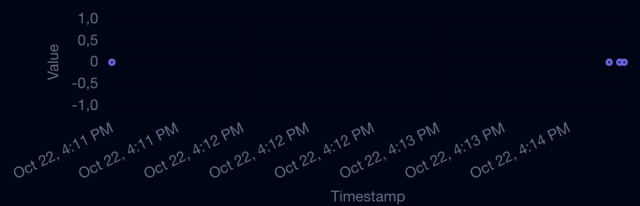
Description

Process data using bash script. Works in all modes (dev/staging/prod) without code changes.

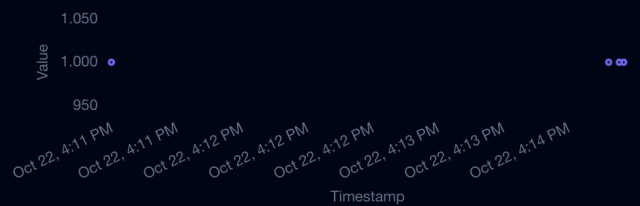
Metadata

Table Plots

custom_messages_count



rows



Lineage

View in graph

Upstream assets

–

Downstream assets

aggregate_results

Definition

Group

shell_example

Code location

dagster_slurm_example

Loaded 5 minutes ago

Source code

Open simple_assets.py in GitHub

Automation details

Sensors

default_automati...condition_sensor

Compute details

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The EUROCC AI Hackathon https://www.openhackathons.org/s/siteevent/a0CUP000013Tp8f2AC/se000375

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